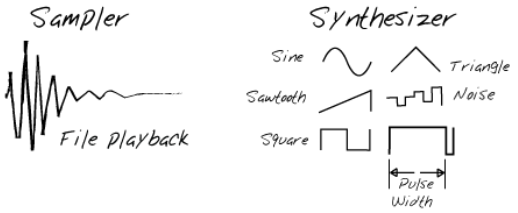
OSCILLATOR (VCO)

Voltage Controlled Oscillator

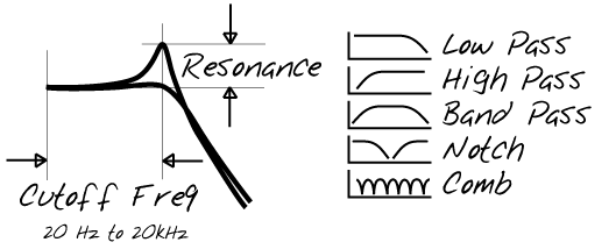
Create the sound



FILTER (VCF)

Voltage Controlled Filter

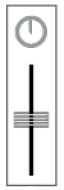
Shape the spectrum



AMPLIFIER (VCA)

Voltage Controlled Amplifier

Shape the volume



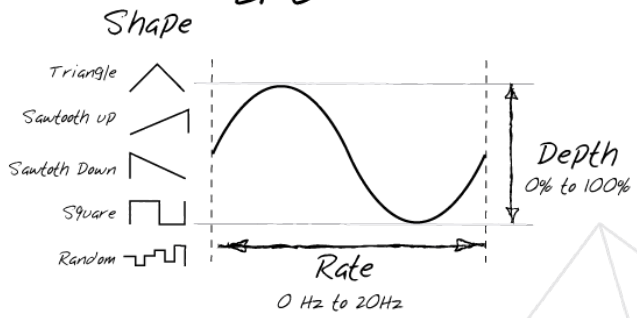
Pitch Sample Select Sample/Start/Phase Cutoff Resonance Pan Volume

Primary Modulation

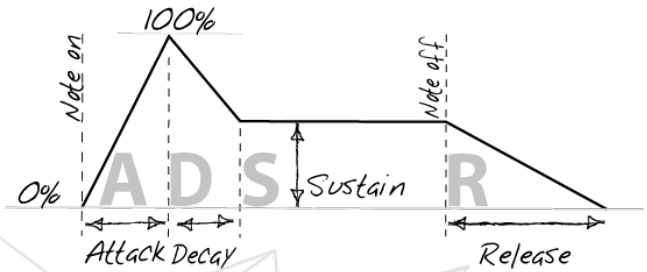
Direct Control

- Key
- velocity
- PitchBend
- Aftertouch
- Continuous Control
  - CC1(modulation)
  - CC2(Breath)
  - CC7(volume)
  - CC10(Pan)
  - CC11(Expression)
  - CC72(release)
  - CC73(attack)
  - CC74(Brightness)

LFO

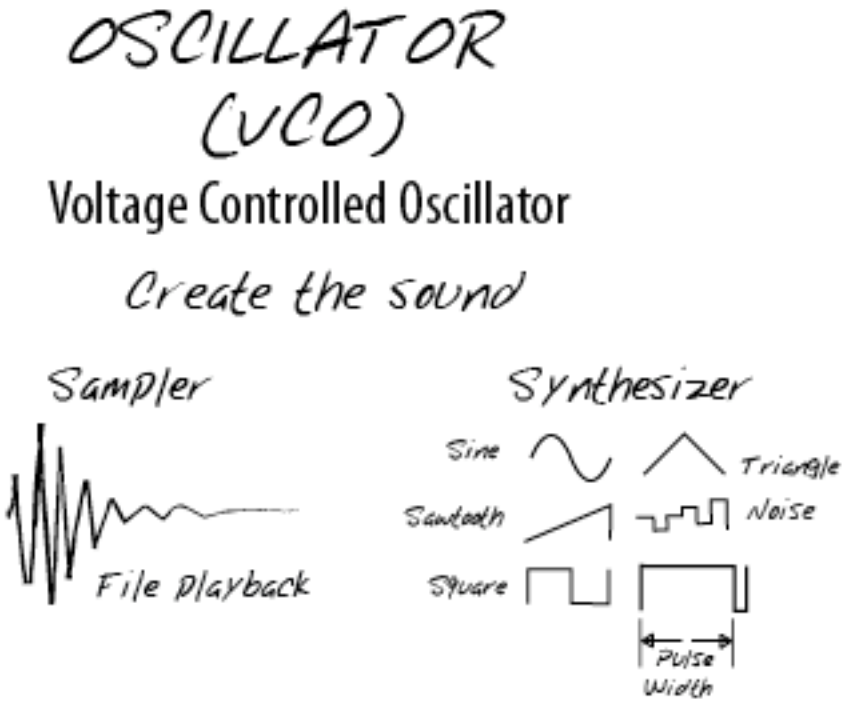


Envelope



Secondary Modulation

Envelope LFO Control



Oscillators are the sound creation portion of a synthesizer. Typically they generate a variety of waveforms such as saw, sine, square, triangle, and noise. In subtractive synthesis a oscillator with a wide spectral content like sawtooth starts the signal then a filter is used to carve out any unwanted portions. Think of the oscillator like marble and the filter a sculptor's chisel. In additive synthesis many simple waveforms(usually sine waves) are added together to create a complex spectrum; they are usually configured in multiples of a common frequency to create a harmonic spectrum. An organ with draw bars is a good example of an additive synth, as you bring in the bars, upper harmonics are added to the overall timbre.

Frequency modulation synthesis also starts with simple waveforms, but this time they are used to modulate(vary) the frequency of each other. When both waves are in the audio range(20hz-20,000 Hz) the results are surprising and tough to predict(at first). Study and experimentation has resulted in some very useful oscillator combinations (algorithms). The end result is the ability to create complex timbres from simple waveforms without the use of filters. It is kind of an advanced additive synthesis, because the beginnings are still simple waveforms and they are being used to create complex timbres. But with FM even two sine wave oscillators can create a wide spectral content, not so with pure additive synthesis. It should be mentioned that FM synthesizers are often very good additive synthesizers as well, because they have many oscillators and the ability to simply add them together.

Register/Octave Semi and Detune: The oscillator section of a synth includes a way to tune the oscillator to the desired pitch. Often there will be 3 knobs, one to choose the octave, one to choose semitones, and a third to fine tune the tuning. These have the most impact when combining oscillators. A small detuning between the oscillators,

around 7 cents or less(a cent is 100th of a semitone), will result in a swirling and chorusy sound. An octave difference is useful for adding weight or shimmer to a patch (think of it like creating a complex timbre with an orchestra). Semitone differences create chords and intervals between the oscillators(good for special effects). A good approach is to set the oscillators to components of the harmonic series;

Harmonic	Note	Oscillator setting
1	root	Unison
2	octave	1 oct
3	octave + fifth	1oct and 7semi
4	2 octaves	2oct,
5	2 octaves + maj 3rd	2oct and 4semi(-13 cents)
6	2 octaves + perfect 5th	2oct and 7semi
7	2 octaves + flat seven(almost)	2oct 10semi (-30 cents)
8	3 octaves	3oct
9	3 octaves plus major 2nd(almost)	3oct 2semi(+4cents)

Tune the cent variations by ear.

Waveforms:
The most common synth waveforms are saw, sine, square, triangle, and noise. Saw is the standard starting point for synth sounds. It has a rich buzzy timbre that includes all harmonics and responds well to drastic filtering. Square is another rich waveform, this one isn't as complex as the sawtooth because it only includes the odd harmonics giving it a hollow timbre. Often with Square waves the synth will give you the option of pulse width modulation. Think of a square wave as a switch going up and down, a typical square wave spends equal amounts of time up and down, but it doesn't need to. Pulse width lets you vary the proportion of the switch being up and down, at 50% they are equal, at 1% it only spends a tiny amount of time up and most of the time down. Pulse width has a dramatic impact on the timbre and it can be modulated with an LFO to create rich undulating textures, but it is also very useful as a static parameter. A square wave with a low pulse width is the best starting point for nasal and vocal sounds(then try pairing it with a formant filter). Triangle wave sounds like a heavily filtered square wave. Like the square it only contains odd harmonics, but it is much less harsh(the upper harmonics are quieter). Triangle is a great choice when you don't have an extra filter but want something more interesting then a sine wave. Sine, the simplest of all waveforms, it is energy at a single frequency. The more complex waveforms can be though of as

collections of hundreds(even thousands) of individual sine waves. A single Sine doesn't respond to filtering at all, because there are no upper harmonics to remove. They do however form the basic building blocks for additive and FM synthesis. Noise comes in a variety of flavors usually denoted by colors, white noise is scientifically useful as it has equal amplitude across all frequencies, but sounds harsh to our ears. Pink noise however, is white noise that has been filtered to the natural characteristics of our ears and is much smoother and generally pleasing. Noise can add a softness to a patch, provide additional high end and make modulated delays like flangers and phasers stand out nicely. Noise is also very useful at the attack portion of the note. In most instruments there is a wide range of inharmonic noise right at the attack, before the instruments settles into the steady state note. Percussion synthesis also relies heavily on noise oscillators. Many synths will include a separate oscillator just for noises, and if you are lucky it will have its own envelope.

Combining Waveforms
When combining waves consider the harmonic series of both waves; a spectrum or FFT analyzer really helps to see what you are creating. One nice combination is to have a square wave combined with a sawtooth that is an octave up. The result is a complete harmonic series, with the two oscillators interlocking like a zipper, they never play the same harmonic! You can then filter and modulate the odd partials(square wave) separately from the even partials(+1oct sawtooth).

Phase and Retrigger
The word phase takes on many meanings depending on context, but in this case phase is a measurement of a point within a single cycle of a repeating waveform. The very beginning of the waveform would be 0% or 0 degrees, half way through the cycle is 50% or 180 degrees, and all the way through the cycle is 100% or 360 %. In analog synths the oscillator is always running, churning out the waveform. This type of oscillator is called free running, when a note is hit it could be at any point in its cycle(any phase), even far away from the zero point. A separate device, the voltage controlled amplifier, acts like a gate turning on and off the sound after it has been created. Because the voltage controlled amplifier takes some time to open up, clicks from starting away from zero are avoided. In modern digital synths we have the ability to restart the waveform at every note, so that the waveform can start precisely where we want it to. Also, modern digital envelopes can be essentially instant, so that click can be dreadfully obvious. There may be phase or retrigger options on your oscillator section, the specifics vary widely, but for percussive/aggressive sounds retriggering the oscillator at a point away from zero adds a nice punch/click to the beginning of the note. On other sounds the click can be distracting and ugly, so use it wisely!

Voices

The word voice takes on a very specific meaning in synth land. If a 4 note chord is held down, that would be 4 voices as far as the synth is concerned. The voices section will limit the number of simultaneous notes; this is done to limit the CPU load and for

creative purposes. There is an overall division between poly(many notes at once) and mono(one note at a time). In mono mode there are additional legato/portamento controls that control how the pitch glides from one note to the next. Voices is not always the same as the number of notes held though. Often synths will have a unison/unisono feature designed to create thick ensemble sounds by playing multiple voices with slightly different qualities(pitch, timbre, onset delay). These typically multiply the number of used voices so if your Unsion is set to 2 and max voices is set to 4, you can only play 2 notes at once before the synth starts “stealing” or stops playing notes.

Standard voices parameters:

Voices, poly, max voices: This sets the maximum number of voices to be played at a single time. 1 often puts the synth into monophonic mode, but you will sometimes see an additional Mono button.

Unison/Unisono: This enables multiple, slightly varied, voices to be used for each played note. The voices are slightly out of tune to create a chorus or celeste sound, the amount of detune is often user configurable.

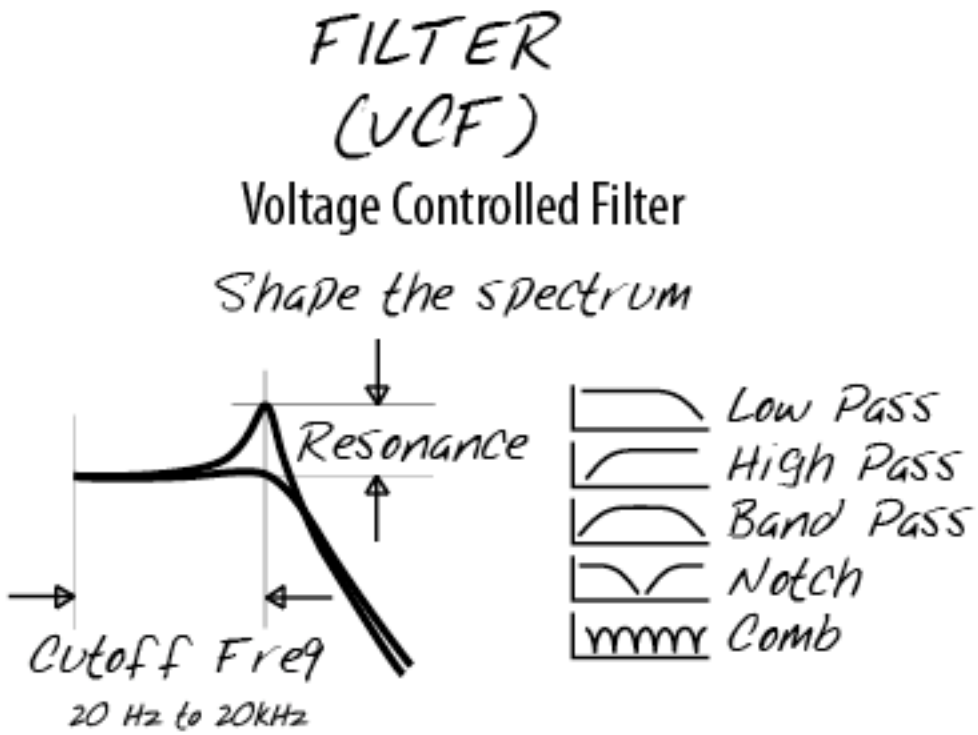
Glide(portamento): On a mono synth(and some poly synths for a very cool sound) the pitch of a note can be made to glide from note to note, like a trombone or violin player can. The amount of time it takes to get from note to note is the glide time(portamento time). Usually glide will need to be enabled and the glide time set. On many synths there will be a legato/glide option near the portamento time control. In legato mode only overlapping notes glide; this modes gives you more control over the gliding and the sound of some notes gliding and some notes jumping is very common in modern electronic music. Set legato mode as your default and get used to playing monosynths that way, you won’t be disappointed.

Priority: This choses which notes are kept active once the maximum voices are reached. The most common setting is Last, which keeps the most recently played note active and starts turning off the oldest notes. Top is also common and can work really well when you are layering a mono and poly synth to highlight the highest note in the voicing; bottom can be used in a similar manner to bring out the bass. On sampled instruments the cutting off of notes above the max polyphony is often referred to as voice stealing.

Important CCs:
There are some standard Control Changes that are important here. Most hardware synths will respond to these, but you may have to program them yourself on a soft synth.

CC5 Portamento time
CC65 Portamento on/off(switch or pedal)
CC68 Legato on/off(switch or pedal)
CC84 Portamento control

Portamento time, portamento on/off, and legato on/off are the same as we have been describing here, but portamento control is special. It doesn't work on all synths, but when it does, it allows you to start a note with a glide without having a note before it; send a value on CC84, the synth treats the value as the pitch of a "phantom" note, the next note played will glide from that phantom note!



Filters are the sound shaping portion of the synthesizer. Usually the oscillator produces a very strong, spectrally intense, sound. To remove the harshness and make it sit well in the mix that brightness needs to be tamed, and that is the role of the filter. A filter is a frequency specific volume control. Every filter description will include both amplitude and frequency information as in “cut the highs.” “Cut” is amplitude reduction, and “the highs” is the frequency description. In this example you would choose a Low Pass filter, the most common synth filter type. Filters are often described by what they let through, the lows are allowed to pass through, the highs are cut. “Cut the lows” is similar, but this time the filter to use would be High pass, the highs are allowed to pass through, the lows are cut. Again, the most common synth filter is low pass, because it removes the excessive high end present in the geometric waveforms created by the oscillator section. The next two filters generally found on synths are Bandpass and Notch. A bandpass filter boosts a specific range of frequencies while cutting above and below that range(band). A notch removes a range of frequencies(band of frequencies). An interesting type of filter found on more endowed synthesizers is comb which adds an evenly spaced series of deep notches. Comb filtering is created with a short delay, and offers the first hint of the peculiar relationship between delays and filters. This type of filter is great at thinning out a complex waveform without loosing the high end grit or bottom end oomph, and it isn’t as obvious as a single notch. Put a comb filter in motion and you are left with a variety of wonderful swirly textures, leave it in place to simulate the complex interactions of a small resonant cavity(the body of a guitar perhaps).

Cutoff Frequency
Cutoff frequency is the most important parameter on any filter, it defines at which frequency the cut or boost is starting. “Cut the highs” is a good start with filter terminology but you will need to get more specific to really craft a quality synth sound. At what frequency does the cut start? That is the cutoff frequency. Say you have a sawtooth wave playing at 261 Hz(roughly middle C), it is creating harmonics all the way up to the top of human hearing(20,000 Hz) and beyond. The sound is buzzy and annoying. The answer is to cut the highs, so add a low pass filter. With the cutoff freq all the way up there will be little if any difference in the sound, it is only cutting frequencies that you can’t perceive(they are too high). As the cutoff frequency is reduced progressively more of the high end is cut, and there is a duller sound. If the filter gets very low the synth will get completely quiet, as all the sound is well above the cutoff frequency.

Resonance
Resonance is one of the most important concepts in all of music, and something that you should study in depth, but like many words it takes on different meanings depending on context. When talking about low and high pass filters, resonance is a boost at the cutoff frequency. A high resonance will dramatically boost harmonics that are right at the cutoff frequency, and then moving the cutoff frequency around will bring harmonics up and down in a beautiful cascading series. The higher the resonance the more pronounced the effect, up to a point, eventually the filter will start to oscillate. When this happens the filter acts like an additional sine wave oscillator(a very loud one). In digital systems self oscillation often sounds horrible, and sometimes sounds good. In analog systems it can be a beautiful thing, you be the judge. On notch and band pass filters resonance acts a little differently, it becomes a width control. The higher the resonance(sometimes known as Q) the narrower the band of frequencies boosted (bandpass) or cut(Notch). A Low resonance notch filter is a secret weapon of synthesist, particularly on big bass sounds; it removes a large amount of midrange energy, leaving room for the other instruments, but still retains the brightness to cut through the mix and the deepness to move the club woofers.

Filter Diagrams
Filter diagrams are also known as frequency response charts. They show frequency on the X axis and amplitude on the Y axis. A reduction in the cutoff frequency is represented by the filter shape moving to the left.

Filter Slope
The slope(steepness) of a low pass filter is described by how much amplitude is reduced for every octave above the cutoff frequency(decibels per octave). The standard two slopes are 12dB per octave(2 pole) and 24dB per octave(4 pole). The term “pole” comes from the mathematical representation of filters; the important thing to know is that every pole represents 6 dB per octave and as poles are added so are the the number of dBper octave; a 2 pole filter is 12dB per octave 3 poles is 18 dB per

octave and so on. The more poles or the more dB per octave the steeper the filter and the deeper the cut close to the cutoff frequency.

Important CCs:
CC74 Brightness typically controls cutoff frequency of a low pass filter.

Filter models
An ideal low pass filter removes all frequencies above the cutoff, but that is impossible in the real world. Designing a filter is an exercise in tradeoffs; designing a steep filter can result in distortion near the cutoff frequency or be heavily processor intensive. There are many circuit designs and many digital algorithms to achieve filtering, each with its own balance of steepness, distortion, cost and features. These tradeoffs and design choices are one of the reasons why some synths sound different/better then others. Modern digital synths offer multiple models for each filter type, the differences are entirely a matter of taste, so experiment and see what you think. The variation between filter models becomes more obvious with high resonance settings, and it can be quite dramatic. The obvious difference between filter models is the amount of distortion added by the filter. To really test this, start with a sine wave and put it through the filter, some models will add additional harmonics to the sine, that is the distortion added by the filter(check it with a spectrum analyzer if you can). Distortion isn't always a bad thing, it can add richness to a simple waveform and really helps to tame extreme resonance, creating a more analog sound(try a ladder filter model for a Moog sounding filter). In some synths there will be a Drive section in the filter which gives direct control of the amount of distortion added by the filter, try controlling that with velocity to give accents a very different character. Or modulate it with an envelope to add bite to the attack of the note. When using high drive settings, it can become necessary to further filter the sound, so put the filters in a serial configuration and add the drive to the first filter, or add an EQ after the synth itself.

Distortion/Drive:
Distortion is non-linear, which in this case means it's effect on the sound is dependent on volume---the louder the input the more distorted the output. Usually, drive will be mostly **harmonic distortion**--- the frequencies present at the output are all harmonics of frequencies present in the input. A sine wave input can result in a complex output(the louder the input the more complex the output), but every frequency present in the output will be a harmonic of that sine wave input. Putting a highly dynamic signal into a distortion will make the loud moments brighter(as upper harmonics are added), and the quiet notes will remain unaffected; **distortion turns volume variations into timbre variations**. You will also notice that the overall dynamic range of the signal is reduced --- **distortion compresses the overall dynamic range**. This dynamic range compression is very useful with highly resonant filters. A resonant filter will cause an incredible amount of volume at a single frequency, often overloading the software/ circuitry and becoming the only frequency that is audible. By adding in software that

distorts the signal above a specific frequency, or using circuitry to do the same, resonant filters become much more usable.

SVF filters
State **V**ariable **F**ilters are very flexible filters with the ability to produce multiple types of filtering simultaneously, and sometimes the ability to morph between them with a single knob. The filter morphing, if your synth offers it, acts like many filters working in parallel, virtually increasing the number of filters available in your synth. Remember that filter design is an exercise in tradeoffs, so with the added ability to morph between filter types you can expect to loose out other ways, CPU usage will go up, and the filter may not have that smooth analog sound that you want, but try it out and see if it meets your needs. The interesting part of the SVF filter is putting that morph control in motion. Modulating morph is very touchy with regard to range and value, start with small amounts of modulation, changing from a low pass to high pass filter is a very drastic change, so small amounts of modulation can add a significant amount of life to the patch. No then, if subtlety is not what you are after

Filter Routing

The typical synth has two filters that can be in a variety of modes(BP,LP,HP,Notch). How the sound moves from the oscillator to the filters, and from filter to filter varies greatly. Now we will examine some common filtering schemes and strategies:

Serial filters. In this organization the outputs of the oscillators are mixed together then sent to the first filter, the output of that filter goes to the second filter. This is a serial (series) connection. A series connection is like bucket brigade, the output of one object is handed to the next object, processed, and handed to the next object and so on down the line. This is probably the simplest configuration to understand, and it is very useful.

Notch and Low Pass
Try setting filter one to notch and filter two to low pass; the notch removes a portion of the mids, but leaves the highs for presence in the mix, and the low pass can be adjusted to control the amount of highs. 12dB per oct on the second filter is good, and play with resonance on the notch to adjust how much of the mids need to be removed.

Source and resonator
Set the first filter keytracking, so it controls the basic timbre of the oscillator up and down the keyboard, but the second filter with no keytracking, it is acting like the body of an instrument(resonator) which doesn't change. This allows you to create formant regions in the instrument, and control the volume of the high notes; as the notes get up into the range of the low pass filter they will naturally get quieter.

Flexible Band Pass
Set the first filter to high pass with a low cutoff frequency, and the second filter to low pass with a high cutoff frequency. This creates a very flexible band pass filter. By adjusting the cutoff frequencies of the two filters you have great control of the width of the filter. Higher resonance settings gives you “horns” or boosts at both cutoff frequencies which can lead to some very interesting results particularly when they are in motion.

Extreme Filter
Configure the two filters exactly the same. If both filters are 24 dBper octave you end up with a 48 dB per octave cutoff! Some synths even include a slave feature on the two filters to make this setup even easier by making the controls of filter two relative to those on filter one, and sometimes all modulation on filter 1 is automatically applied to filter 2, very handy if it is an optio. It can be interesting to begin with the filters exactly the same, then give them slightly different modulation.

Dual Notch
If the synth has filters with notch mode, put them in motion. Slow moving notch filters sound great on soft pads. Try using 2 LFO’s, one per filter, and move them slow and wide. Set the frequencies of the LFO’s different so that the texture doesn’t loop predictably. Add in a little bit of noise going to both filters and unison mode with as much polyphony as you can manage and you have a great pad on your hands. Just use a little noise though, remember that it builds up with every note, so what sounds like a nice amount of analog warmth on a single note becomes an annoying hiss as the notes are stacked up.

Bring out the noise
The tough part about the serial configuration is when something is cut with the first filter, it is gone and can’t be brought up with the second filter; or can it? Try using a 24 dBper oct low pass filter, set quite low, so only the first couple partials come through well. Follow that with a band pass filter with a high cutoff frequency. At first you won’t hear a difference, the harmonic content up there has been dramatically cut by the low pass filter. As you raise the resonance of the band pass filter you can bring some of those harmonics back up. The resonance will have to be quite high to make this work, leading to a very narrow bandwidth, only a couple harmonics will jump up. And, since these harmonics have first been filtered heavily then brought back up, they will be very colored by the synth. On an analog synth you will be bringing up noise along with the signal which is not always a bad thing(sometimes noise equals character). On a digital synth, where noise is not nearly an issue, you will be exposing the quality of the digital filters. So, this would be a great time to try out those different filter models to see how they vary.

Notch and Band Pass

A similar approach can be taken with a notch filter and a band pass filter at the same cutoff frequency. The two fight each other in a constant tug of war, the notch pulling down while the band pass pulls up. One would think that the two would cancel out, but filters also introduce phase variations which adds richness to the sound. Try modulating the two filters differently, and experiment with different resonance settings. A low resonance on the notch and high on the BP is particularly cool; the result is a general cut of the mids with a spike of resonance that you can place wherever you need it in the mix by adjusting the BP cutoff frequency.

Parallel filters. In this organization the output of the oscillators is mixed, then split and sent to the individual filters, then the output of the filters is combined. The filter outputs are summed(mixed) before the amplifier, thus a single envelope controls the overall amplitude. Some synths will have a separate amplitude envelope for the filter outputs, a nice feature indeed.

Dual Low Pass

There is a smoothness with parallel filters that isn't there with serial filters. Set both filters to low pass mode(24 dBper oct). Set the cutoff frequencies for both around 300Hz, then modulate one of them slowly. The beauty of the parallel configuration is the output of the second filter only ever adds to the output of the first. In serial configuration the second filter cuts the output of the first resulting in very steep slopes and some drastic filtering. The parallel configuration is softer and smoother, when the second filter opens up it adds to the texture, and if the second filter goes all the way down to 0Hz, the first filter output is still strong. This smoothness makes the parallel configuration great for undulating pads.

Flexible Notch with horns

One High Pass filter and one Low Pass filter. Set the cutoff frequency of the high pass well above the low pass creating a deep notch in the midrange. With a strong low end, and a strong high end this configuration can make for an aggressive patch. A powerful bass sound can be made this way, but because of the high pass filter it can be a harsh configuration; following the synth with a high shelving EQ or low pass filter helps with the harshness. Another option to control the harshness is to bring the volume down on the input to the high pass filter. Some synths offer a flexible crossfade that controls how much signal to send to the two filters, by setting it to send 90% to the low pass and only 10% to the high pass the volume of the high end is greatly reduced.

The width of the notch is controlled by the individual cutoff frequencies, raise the high pass or lower the low pass to widen the notch. The slope of the filters controls the depth of the notch; 12dB per oct filters will create a gentle midrange dip, where 24dB per oct will create a deep midrange cut (this interacts with the

cutoff frequencies also, an FFT analyzer really helps to see as well as hear what is going on). By modulating the high pass filter the sound can take on high end complexity while leaving the lower partials strong and consistent (particularly good for bass sounds).

With the high pass filter, try inverting the envelope(or putting the modulation in the negative direction). The envelope will shoot the filter down then raise back up during the decay phase(set sustain to 0). The result is an attack with a strong midrange component, and the notch widens as the note is held, and unusual but musically interesting sound. Try routing a little noise to the high pass filter to add some warmth to the high end and bring attention to any filter movement. A bit of noise makes filter movement more obvious without getting the obvious jumping of individual harmonics, particularly with higher resonance settings.

Twin(or more) Peaks

Both filters are set to Band Pass mode. While there are many ways to use this flexible configuration, start with one of the filters centered around the fundamental. Modulate cutoff frequency with key position to keep the timbre consistent across the keyboard. Set the second band pass filter with a higher cutoff frequency and resonance and set it to keytrack as well, but not exactly with key position. Experiment with the amount of keytracking on the second filter. Every acoustic instrument changes quality across its range, some quite dramatically. By altering keytracking amount the patch will have areas of the keyboard that sound different then others, this is particularly true with higher resonance settings, where certain notes will really jump out(when the cutoff frequency is centered over a harmonic). Each note takes on a unique but consistent character. The resonance of the first filter will control how many low harmonics get through(a high resonance results in the fundamental dominating). The resonance and cutoff of the second filter will control the upper harmonics. A good start is a low resonance on the second filter to get a broad upper harmonic content, but that is not the only approach. High resonance settings and careful cutoff frequency settings for both filters can create strong vocal formants, a third parallel band pass filter can take the vocal formant filter concept even further. For interesting talk-box and wah effects add CC control of cutoff frequency with different amounts and possibly direction for the two filters.

Lowpass and Band Pass

Set the Lowpass filter keytracking near the fundamental to create the basic timbre of the sound. Very little to no keytracking on the bandpass filter. Set this filter's frequency and resonance with the whole mix going as a way to fill in a hole in the mix you will be sure it can really cut through when necessary. Upper mid range right above the vocalist is a good place to start. If possible modulate the band pass filter resonance with an envelope, then set velocity to modulate envelope amount. Set the envelope with no attack, sustain, or release, then adjust decay so you get a bit of bite at the beginning of each note. This envelop will cause added emphasis at the beginning of each note, and the velocity control

will be great for accents. Because the filter is not keytracking the accented frequencies will be stationary in the mix and easy to control. A hint of overdrive or amp simulation after the synth will bring out the accents even more. Like the Flexible Notch be careful with the volume of the higher filter. If you can, send more signal to the lower filter and less to the higher one, otherwise use an EQ after the synth to control the harshness.

Filter Thunk

Many instruments exhibit a low frequency thunk right at the attack of the note, piano and guitar are notable examples. A resonant filter with a low cutoff frequency (100Hz or lower) will vibrate sympathetically with the oscillator creating a low thunk with every note. The amount and quality of the thunk is very different from synth to synth, so try it out on a couple and see what you think. As strange as it may seem, the amount of resonance will control the duration of the low freq thunk(another hint at the relationship between filters and delays: filter resonance = delay feedback). Give the filter a little bit of keytracking so it changes across the keyboard, but not too much. Then give it quite a bit of velocity modulation, so hard notes include quite a bit of the thunk, and light hits very little. Then use less overall, this should be a subtle effect, don't push it too hard or the element could easily muddy up the bottom end of your mix. In fact, you may want to add a low cut EQ after the synth to remove some of the thunk.

AMPLIFIER
(VCA)
Voltage Controlled Amplifier

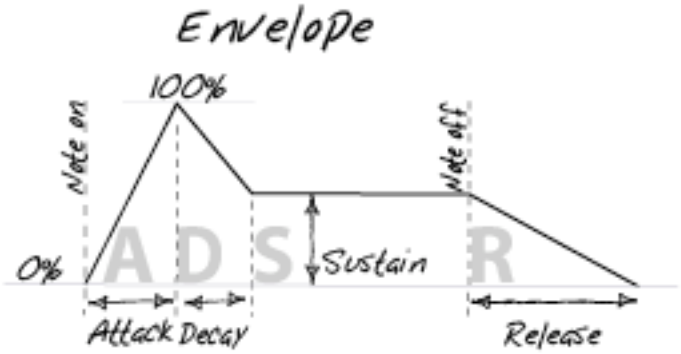
Shape the volume



The final stage(before the delay effects) is an amplifier(volume control). This is almost always modulated with an envelope and creates the volume contour of the individual notes. To really understand the Amplifier you must first learn about an envelope.

Envelopes are general purpose modulators, and can be sent to a variety of destinations, but the most important one is the Amplifier envelope. The typical envelope is divided into 4 stages.

Attack, Decay, Sustain, Release---ADSR for short



Attack, Decay, and Release are all amounts of time(milliseconds usually), Sustain is different, it is a level.

The envelope starts at zero with a MIDI note on event and proceeds to full level over the **attack time**; then it continues to the **sustain level** over the **decay time**; it stays at the **sustain level** until a MIDI note off event is received, at which time the envelope returns to zero over the **release time**.

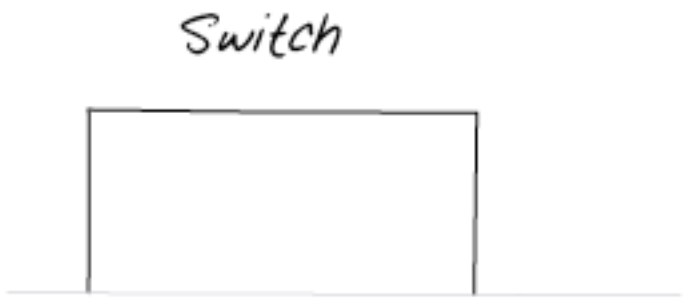
Again, an envelope can be routed to a variety of parameters in a synthesizer, but for now we will be considering the amplitude envelope, the most important envelope.

Sustaining and Nonsustaining sounds:
All instrumental sounds can be placed in one of two categories, sustaining, or non-sustaining. If energy is being added to the instrument over the course of a note (blowing, bowing, mechanically vibrating) then it is a **sustaining instrument**. If the instrument gets an initial burst of energy then is left to resonate and decay(striking or plucking) then it is a **non-sustaining instrument**.

To emulate **sustaining** instrumental sounds, the sustain portion of your amplitude envelope is non-zero.

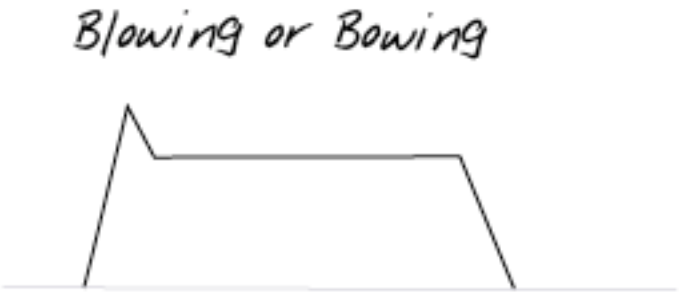
To emulate **non-sustaining** instruments, the sustain portion of the amplitude envelope is zero.

Some common settings for the amp envelope:



Attack 0, **D**ecay N/A, **S**ustain 100%, **R**elease 0.

Switches are on or off with nearly instant movement. So, it takes no time to go from zero to full volume, attack is 0. Sustain is at 100%, so decay does nothing, there isn't a different level to go down too. On Note off the sound should stop instantly, so Release is set to 0. Organs can be emulated with this envelope. *To add a little bite to the beginning of the note reduce sustain to 80% and set decay very short. That is good for percussive organs.*

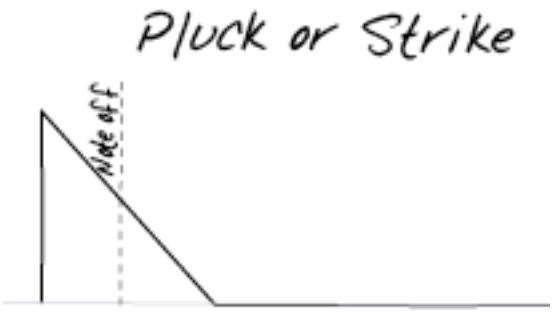


Attack 0-10%(almost 0), **D**ecay 0-10%, **S**ustain 50-90%, **R**elease 0-10%.

These percentages are rough guides and will vary from synth to synth. But the idea is when the player starts to blow or bow there is an initial strong burst of energy, and then the note settles into a steady sustaining level. That initial energy causes the envelope to jump from 0 to full volume quickly(the low attack time setting) then quickly down to the sustain level(the low decay time setting). The note holds at the sustain level---any variations in level during the sustain are added with LFO's--- until the note off, where the instrument vibrates for a moment, trailing off after the player has stopped blowing or bowing.

Increasing attack time and decay time will soften the initial transient. Increasing sustain level will give more power to the note and bring it to the forefront of your mix. Increasing release time sounds a bit like adding reverb and can help to smooth out the lines when moving from note to note.

This envelope is the standard sustaining envelope and is adjusted according to the actual instrument. For strings the attack decay and release would be longer---but this varies according to articulation. For horns the sustain level would be lower as the initial burst of air is quite strong when compared with the sustaining level ---again this varies according to playing style.



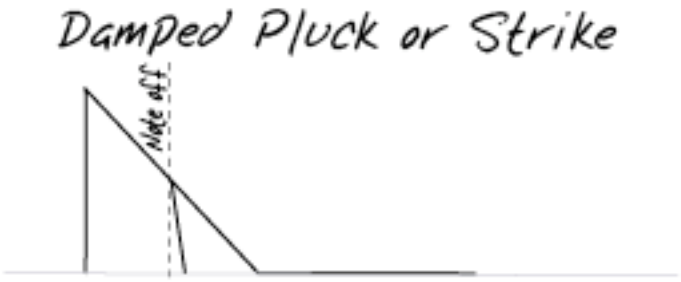
Attack 0, **D**ecay 10-80%, **S**ustain 0, **R**elease same as decay.

This is a non-sustaining envelope that plays out regardless of how long you hit the note for; perfect for emulating struck or plucked instruments. When an instrument is struck or plucked it goes to full volume nearly instantly, then decays based on the decay time.

Release is set similar to decay so that if the player releases the MIDI note the envelope will continue to play out naturally. The Decay(and release) parameters are adjusted based on the instrument. A Gong would have a very long Decay, while a marimba note would be very short.

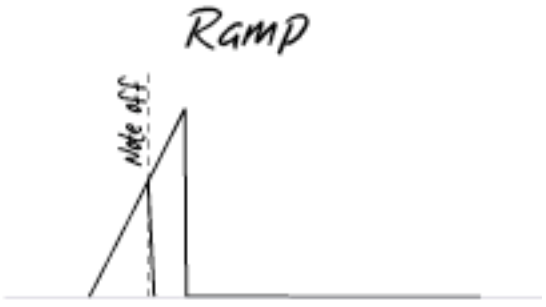
Envelope trigger mode:
Some synthesizers give a variety of modes for envelope playback, one possible mode is Trigger or One-Shot. in this mode the envelope plays out completely, ignoring the note off, and is particularly useful for percussion sounds.

Keytracking and envelope time:
Often higher notes decay faster then lower notes in real instruments, to make that happen on your synthesizer you would need to route keytracking to decay and release time(negatively, because high notes need to cause short values)--this is a common parameter on synthesizers. Some synthesizers have a general purpose time control for the envelope, routing keytracking to control this parameter has a similar result.



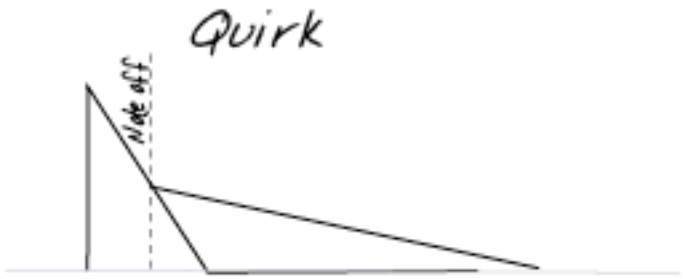
Attack 0, **D**ecay 10-80%, **S**ustain 0, **R**elease 0

This is very similar to the pluck or strike, but because release time is zero when the envelope receives a note off it goes to zero, stopping the sound. This is representative of a piano, when the key is pressed a felt hammer strikes a string, and when the key is released a damper raises up and stops the note. A similar envelope would be used to emulate electric guitar and bass.



Attack 80-100%, **D**ecay 0-20%, **S**ustain 0, **R**elease 0

This envelope emulated a reversed percussion sound, it swells in and then suddenly stops. It is great for transitions to build energy, but there really isn't a real instrument with this type of envelope.



Attack 0, **D**ecay 10-30%, **S**ustain 0, **R**elease 50-100%

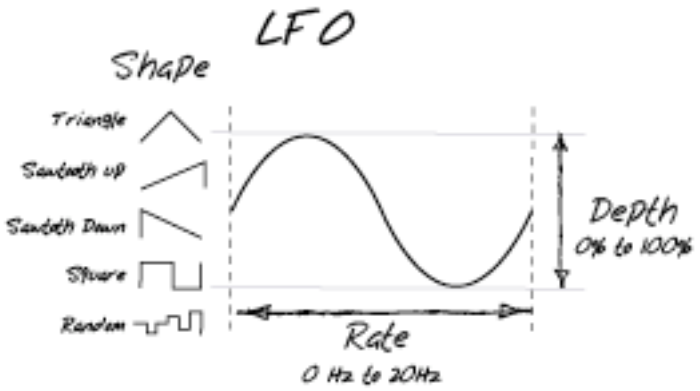
This is a funny envelope, not really that useful, but interesting to try. With these settings if you hold a note down for a long time the envelope goes through its short decay phase and you end up with a short percussive note, like a pluck. But if you play a really short MIDI note, one where the note off happens within the decay time, then the release phase kicks in and you get a long audible result. Long MIDI notes end up with short results, and short MIDI notes end up with long results!

Important CCs:
CC7 Volume
CC11 Expression
CC72 Release Time
CC73 Attack Time

CC7 and 11 both control volume, the standard usage is 11 for performance dynamic variations, and CC7 for mixing.

Panning and Keytracking:
The amplifier section in a synthesizer often includes panning. This common mixing tool can be put to creative use as a modulation destination. Routing keytracking to panning is useful for creating piano-like panning, where the high notes are on the right and low notes on the left.

Panning and per-note random:
Another common usage of panning is slight on note randomization. Giving each note a little bit of random panning increases the width of the instrument without adding a swirling or chorus-like movement.



A **L**ow **F**requency **O**scillator, LFO for short, adds steady cyclic motion to a synth parameter. While it is a general purpose modulator that can control a variety of parameters, creating a vibrato effect is its most common usage---vibrato is a steady variation in pitch(use a semitone for depth and 5Hz for rate as a good starting point for natural human vibrato). To create vibrato the LFO's destination must be set to the frequency of the oscillator. This particular modulation is so important many synthesizers have a dedicated LFO specifically for vibrato with and internally set to a sine or triangle shape and a destination of oscillator frequency. The LFO shares many of its important parameters with the oscillator: shape, rate(frequency). Notice the range of the rate parameter; an oscillator runs at audible frequencies(20Hz to 20,000 Hz), where an LFO runs at much lower frequencies(0Hz to 20Hz--- though many synthesizer's LFO's can run faster for interesting effects).

Sync:
Most LFOs can be set to synchronize with the global song tempo. When in sync mode the Rate parameter will be set in metric subdivisions(eighth note, quarter note, etc...). This is great for making rhythmically accurate pulses and timbre variations. Use triplet values to keep a sense of sync but loose the rigid quality of eighth and sixteenth note rates.

Retrigger or Free:
An LFO in retrigger mode will start from the beginning of its shape at the MIDI note on. This is perfect for keeping consistency between notes. With retrigger off(free mode), the LFO keeps running even when no notes are playing, so each note starts at a different point in the LFO cycle. This will add note to note variations and an unpredictability to your patch.

An important consideration of retrigger off or free is in playing chords. With retrigger off, all notes that are played will have perfectly in sync LFO movements---it is like there is a single LFO controlling all the voices in perfect unison(perfect for synced LFOs as they will always be right in time with the global meter).

With retrigger on, the LFO restarts with each note, so a chord played with staggered attacks would also have staggered LFO movements--- it is like there is an LFO

individual to each voice and it starts over at the note on. The vibrato in a group of players is never in perfect sync, so it makes sense to have individual LFOs to emulate their vibrato.

If the LFO is in sync mode the rate will follow the global tempo, but the LFO will restart with each note, so depending when the note is hit the LFO could be consistently ahead or behind the beat.

Keytracking and LFO rate:
The vibrato example can be taken a bit further by adding a secondary modulation to LFO rate. Even with retrigger on, if all notes of a chord are played simultaneously the vibrato of the individual notes will be in sync. To avoid that, route keytracking to LFO rate, every note on the keyboard will then have a different vibrato rate, and there is no way for the LFOs to sync up perfectly. Sending velocity to LFO rate has a similar effect. For pad and choir sounds this really helps thicken up the patch and individualize the voices.

LFO attack:
Back to the vibrato example; often vibrato is absent during the attack of a note then gets more dramatic as the note sustains---this is emulated in synthesizers by controlling LFO depth with an envelope. Instead of adding an entire ADSR envelope it is a simple ramp up to full level. The time it takes for the LFO depth to reach maximum is set with the LFO attack parameter.

Delay:

Most modern synths include an effect section after the amplifier, while it may contain spectral effects like filters and EQ's, and dynamic effects like compression and limiting, the most useful(and common) addition is a flexible delay section. Usually it will be a stereo delay with a wide range of delay times and syncing capability.

It is hard to overstate the importance of Delay in synthesis. A simple delay can turn a boring synth part into a thick and captivating soundscape. The surprising part about delays is the huge variety of effects it is capable of creating. Comb filtering, stereo width enhancement, doubling, acoustic simulation, and impossible echos are all delay effects. In fact, many of the plug-ins under a variety of names are all delay effects with slight variations(phaser, flanger, delay, echo, reverb, chorus).

Delays will always have 3 main controls: Delay time, Feedback, and Dry/Wet(mix).

Delay time is measured in Milliseconds or seconds and is the time between the original sound and the first echo caused by the device.

Feedback controls how many times the delay happens, or more specifically how much of the delayed signal is routed back into the input of the device(feed back to the input). At 0% feedback a single echo is heard, at 100% feedback a single impulse at the input goes on forever, constantly recycling through the device(above 100% causes the volume to increase with every repeat which can lead to uncontrollably loud signals very fast, watch out!). With the short delays, feedback will increase the impact of the effect from subtle to extreme(often metallic and harsh, and sometimes introducing extra pitches into the signal that are dependent on the delay time).

Dry/Wet(Mix) is a blending between the original signal and the delayed signal, at 0% only the original is heard at 100% only the delayed signal is heard.

Delay time LFO modulation is the most common addition to delay effects. By varying delay time with an LFO the delay comes to life. On short delay effects like Chorus, Flanger, and Phaser, the LFO is essential to the swirly/spacey character of the effect. There will be the usual LFO controls, Depth, Rate, and Shape. For most situations subtlety is rewarded, so keep the depth low. To get more extreme with the LFO consider a high depth with a very low rate to introduce slow sweeping sonic changes(particularly useful with a phaser or flanger), or use a low depth with a high rate(this adds a nice roughness without being overpowering).

Delay time envelope follower modulation an envelope follower listens to the incoming signal, calculates the average amplitude and uses that as a control signal to change another parameter, delay time in this case. With a phaser the results can seem like an auto-wah guitar effect, with each amplitude jump in the incoming signal the delay

time changes and shifts the comb filter around. On flangers the shimmery flanged quality shifts with each hit, very effective in bringing hat parts to live. When using envelope followers many parameters must be set correctly to get an appropriate sound (initial delay time, envelope threshold and rate, and envelope depth). So, tweak carefully and slowly, and consider the depth and direction of the envelope's motion.

Comb Filter:
This is where the curious relationship between filter and delay really starts to show itself. By mixing a signal with a delayed copy of itself some frequencies cancel out and others reinforce, and the frequencies of reinforcement are always in an harmonic series. When viewed on a spectrum analyzer these evenly spaced notches resemble the teeth of a comb, hence the name comb filter. A short delay is a complex filter!

The sound of comb filtering is all around us, everywhere you hear the same sound reflected off of two surfaces, there is comb filtering. We are unconsciously aware of the effect and use it every day to understand our surroundings; every room has a particular sound caused by the relative distance of all the walls to you and to each other, every room has its own set of comb filters.

Comb filtering is particularly evident with broadband noise(which has energy at all frequencies) and when the delay time is changing(the sound source is in movement, but the reflective surface is stationary). Experiment for yourself, create a wind noise with your mouth and move your self forward and back from a wall, listen carefully. As you move closer to the wall, between 2 feet and 2 inches, the tonality of the noise changes. The direct sound of your mouth is mixing with the reflected sound from the wall, but because sound takes time to move through the air there is a delay between the two. The direct sound from your mouth reaches your ear a tiny bit earlier then the reflected sound from the wall. As you move, that delay changes and the comb filter shifts causing an observable swooshing quality to the noise(flanging).

When the combined sound reaches your ear there are specific frequencies where direct and reflected sound are "out of phase" and cancel each other out. The direct sound is in the compression phase(positive pressure or amplitude) while the reflected sound is in the rarefaction phase(negative pressure or amplitude). There are also frequencies where the combined sounds are in phase and reinforce each other. The areas of reinforcement are located in a harmonic series.

Delay time and frequency are both measurements of time, actually reciprocals of each other. Delay time is "seconds per event", and frequency is "events per second". A delay time of 1 millisecond is equivalent to a frequency of 1 kilohertz(pretty close to a "C" two octaves above middle C). So if the delay in sound is 1 millisecond(a thousandth of a second) then the resultant comb filter will have peaks in a harmonic series based on 1 kilohertz. As the delay time goes down the fundamental frequency of the comb filter goes up.

Problems with Comb Filtering when Recording:

Comb filtering is a constant consideration when recording. A comb filter is the sound of the space and possibly the sound of the recording equipment(small delays happen all over the place!). For most recordings the goal is to recreate the actual sound of the instrument, to make the gear used and recording situation invisible. But when there are two mics recording, there will be comb filtering. When there is a music stand or monitor or podium in front of a vocalist, there will be comb filtering. When recording both a direct signal and a mic'd cabined, there will be comb filtering. In all these cases the comb filtering weakens the sound and pushes it into the background of a mix which can be artistically useful, but in most cases it is something to avoid.

Depending on the situation there are a variety of ways to correct it, but the most important thing is that you HEAR it. The earlier comb filtering is recognized the easier it is to correct. Listen carefully while preparing for any recording. As mentioned earlier walls, podiums, music stands, any flat surface really, can cause comb filtering. If you recognize the issue before recording, make adjustments before recording. Move the mic further away from the wall. Put the music stand off to the side. Get rid of the podium. Change the height of the mic. Anything to get rid of the phasing(another term for a comb filter). If there is a phase issue in a single microphone it is pretty much impossible to fix later, EQ can help, but really those frequencies are lost for good.

If the phase issue is caused by using two microphones at different distances, that can be fixed in the DAW later. On the mic that was closer to the source nudge the audio later till it lines up perfectly with the other recording. Zooming way in on the waveform view in your DAW makes this very easy. This will need to be done with every recording and can become tedious, so there is another solution.

Add a delay to the closer mic channel. Many DAW's include a sample delay, a plug in designed to create a very short delay in the signal flow, and it is a very useful device. Plug it in to a track and change delay time till the combined sound is best, you will feel it lock into place if you listen closely. A correlation meter can help, to use it pan the two recording hard right and left and adjust your delay till the correlation meter reads highest for the longest amount of time. Trust your ears more then your eyes. Once you have the delay set, don't move the mic. Move the mic even a little bit and you will need to readjust the delay.

Comb filtering is particularly problematic on the bottom end. When recording a direct bass and a mic'd cabinet the comb filtering can simply destroy the bass tone. Line up the two tracks by nudging or adding delay and wow, the result is amazingly tight solid bottom end.

All that is well and good for simple sources like a vocal or bass where the sound comes from a single location, but what about a complex sound source like a drum kit or piano where the sound is coming from a variety of places? These are the toughest to record, there will be phase issues, no matter what you do. Placing mics on a complex source is

a balancing act, a trade-off. Let your ears be your guide, and remember that phase problems are much more noticeable in the low frequencies, so pay careful attention there.

Creative uses of comb filtering:

The undesirable effect of “thinning” caused by comb filtering can be just what is needed to fit a spectrally rich element into a mix. Often an element is too big, it dominates and is hard to place in a busy mix. Distorted guitars are a prime example because their unnaturally strong mid and upper frequencies compete with everything else in the mix. Comb filtering can be a perfect solution as its many evenly spaced cuts thin out a spectrally rich signal without dulling the high end. Using this on distorted guitars is an excellent way to fit them into your mix, but keep them bright and powerful. Powerful synth sounds can also benefit from a similar treatment.

Flanger, a comb filter in motion:

A Flanger is an extension of the comb filter. It has an internal short delay to cause a comb filter, this delay also includes a variable feedback, which when set high makes the sound ringing, metallic, and gives the effect a particular note(because of the harmonic series relationship of the peaks in a comb filter). The final important feature of a flanger is an LFO controlling the delay time. As the LFO raises and lowers the delay time, the comb filter moves down and up. Like all delay effects there is a dry/wet(mix) control to balance the effect and the unprocessed portion of the sound.

Flanger will be most obvious on noisy, distorted, and spectrally rich content. For the flanger to work well there needs to be sound there to filter! Distorted guitars, high hats, bright/unfiltered synth patches, and synth patches with a strong noise component.

The LFO in a flanger can be used to add motion to static parts. A simple hat loop can get boring after multiple times looping, but a flanger with a slow LFO will cause the filtering to be different with each loop and add life to the static part.

The LFO can also be used to add stereo width to a mono element. A stereo flanger will process the right and left channels separately, pushing the sound out to the sides and leaving the center of the mix empty to place another element(melody maybe!).

The downside of the flanger is how identifiable it is; not a subtle effect. The distinct tonality of the effect limits its effectiveness on tonal material(the note of the flanger can conflict with the note in the source audio). Because of this, the flanger is largely used for special effects and in short bursts. If a less obvious effect is needed for “thinning” and complex filtering, a phaser is often a better tool for the job.

Phaser, a flanger without the harmonic series:

A phaser is much like a Flanger in that you have a complex filter in motion(controlled by an LFO). Unlike the Flanger thought he dips and peaks in the filter are not set in a harmonic series. Phaser designs vary widely, but it is always a complex filter with an LFO. The cuts in a phaser are not set into a harmonic series, and on a better equipt phaser the number and organization of the cuts can be varied opening up a wide range of sonic possibilities. Because the filters are not in a harmonic series, the obvious tonality of the flanger is eliminated.

Phasers, like flangers, are best used on spectrally rich and noisy material. There needs to be sound there for the phaser's cuts to be heard. Put a phaser on white noise and the effect is very obvious as each cut can be heard as it moves through the noise. Put the phaser on a simple(dull) synth patch and the effect will only be recognized when it happens to be aligned with one of the harmonics of the patch. So, to use it effectively will require the user to fine tune the frequency of the cuts to line up with the sound it is filtering.

Again like a flanger, the LFO on a phaser can be used to give static sounds motion and mono sounds stereo presence. For generally useful, subtle effects, set the LFO slow and deep(low rate, high amount) or fast and shallow(high rate, low amount).

Chorus: Create a thick choir-like sound(celeste). By mixing a signal with an out of tune version of itself the impression of an ensemble is created. Modulating the delay time of the delayed version creates pitch variations meant to emulate a multitude of performers. How chorus is implemented varies greatly between manufacturers, but most use multiple delay lines and separate LFO's on each to create the shifting and shimmering texture. This is usually a stereo effect, and can be used to widen a mono signal in a stereo field. A typical application of chorus has feedback set to 0.

Stereo width and delays:
Stereo width is created by treating the left and right channels differently(a mono signal has no width, it is exactly the same in the left and right channel). Some common approaches are detuning the left and right oscillators, filtering differently in the left and right filters, and modulating differently in the left and right channels. But, the most common way to control stereo width is by using short delays differently in the left and right channels. Any of the delay effects(comb filter, phaser, flanger, chorus, and longer delays) can be set differently in the left and right channel, contributing to the perceived width of the synth.

The important parameter here is delay time. A slight difference in delay time between the left and right speaker (below 50ms) will have a panning effect, shifting the perceived location of the sound toward the side with the lower delay time. This psychoacoustic phenomena is useful on its own for stereo localization, but by modulating the delay times the signal can be made to move around in the stereo field, creating a sense of

width. This was mentioned earlier as a typical feature in chorus flanger and phaser effects.

Modulation Strategies:

Modulation is the time dimension of synthesis. The standard Modulators are Direct control(key, velocity, CC, pitchbend, and pressure), Low Frequency Oscillators, and Envelopes. The most common destinations are frequency, filter cutoff, and amplitude. This scheme is extended by adding secondary(modulation of the modulators). So, LFO's get two modulation inputs, rate and amplitude, and envelopes can have each stage(ADSR) modulated.

Here is a list of the standard primary and secondary modulations.

Primary Modulations:		
Modulator	**Parameter**	**Comment**
Key	Pitch	Also called keytracking, this is on by default in most samplers.
Key	Sample Select	Only on samplers with multi sampling. Creates realistic instruments.
Key	Filter Cutoff	Control brightness across keyboard.
Key	Pan	Emulate natural piano panning.
Velocity	Sample Select	Only on samplers with multi sampling. Creates realistic dynamics.
Velocity	Amplitude	Accented notes are louder.
Velocity	Filter Cutoff	Accented notes are brighter.
Velocity	Sample Start	Only on samplers, negative modulation, higher velocity starts the sample earlier, bringing in the natural attack of the sample.
Envelope	Amplitude	Gives the sound its basic shape.
Envelope	Filter Cutoff	Usually used to add bite to the attack of every note. Longer attacks can create auto-wah type effects.

Envelope	Filter Resonance	Usually used to add bite to attack, can be more natural then moving cutoff freq, or can be used in combination with it for complex filter movement.
Envelope	Pitch	Use this to add pitch movement to the attack. Usually starts high and settles quickly down into the pitch.
LFO	Pitch	Vibrato, usually between 2 and 8 HZ and a depth of less then a semitone.
LFO	Amplitude	Tremolo
LFO	Pan	Autopan
LFO	Filter Cutoff	Adds motion and interest to sustained notes.
CC74(Brightness)	Filter Cutoff	Allows for manual control of brigtness.
CC10(Pan)	Pan	Standard mixing pan control.
CC7(Volume)	Amplitude	Standard mixing volume control.
CC 11(Expression)	Amplitude	Acts as a percentage of CC7.
Aftertouch	Filter Cutoff	Used to add modtion and interest to sustained notes.
Pitchbend	Pitch	Adjustable range. Whole step, minor 3rd, and octave are common.
Secondary Modulations:		
Modulator	**Parameter**	**Comment**
CC1(modulation)	Pitch LFO Depth	Manual control of vibrato amount. When using modulation you can set the Pitch LFO depth very high, that way you can add short bursts of extreme vibrato when needed.

CC73(attack)	Amp envelope Attack	Manual control of attack time. This is a great control for switching between legato and staccato passages with one sound.
CC72(release)	Amp envelope Release	Manual control of release time. This is a useful control adjusting the density of a part, release often acts similarly to reverb, reducing it leaves space between notes.
CC	Amp envelope Decay	Manual control of decay time. This is most useful on percussive sounds with no sustain. It will control the length of the sound.
Velocity	Amp envelope Attack	Negative modulation, higher velocity have shorter attack phase.
Velocity	Filter envelope Amount	Higher velocities have greater filter movement, usually makes attack brigher while leaving sustain consistent.
Velocity	Pitch Evelope Amount	Accented notes have larger pitch change.
Envelope	Pitch LFO Depth	Usually used to fade in vibrato with a slow attack. Often a simple envelope is built into the LFO itself as an attack parameter.
Key	Amp envelope Decay	Negative modulation, control the length of percussive notes across the keyboard, higher notes are shorter then low ones.

Often it is the modulation that defines a sound, and it is modulation that makes a sound expressive and dynamic. When creating and exploring synthesizers, the configuration of modulation is the most complicated aspect, and the par that requires the most forethought. Once the modulation is configured the performing and knob tweaking can start. This is particularly true with modular and semi modular synths. With those type of devices configuring a patch with your standard modulations and saving it as a template for future patches is a great practice.